

HIGH LEVEL DESIGN (HLD)

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Project Title: RAG-Based Customer Support Assistant with LangGraph & HITL

1. System Overview

RAG-based assistant that retrieves answers from PDF documents.

2. Architecture

User → Retriever → Generator → Decision → Output/HITL

3. Components

Document Loader, Chunking, Embeddings, ChromaDB, Retriever, Generator, HITL

4. Data Flow

PDF → Chunk → Embed → Store → Query → Retrieve → Generate → Output

5. Technology

Python, LangChain, ChromaDB, HuggingFace, LangGraph

6. Scalability

Supports large documents and can scale with better infrastructure.